CS767 Advanced Machine Learning and Neural Networks

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Homework Assignment 1

1. Machine Learning Paradigms
    a. Supervised learning
    b. Unsupervised learning
    c. Reinforcement learning
    d. No learning
    e. Supervised learning
    f. Unsupervised learning
    g. Reinforcement learning
    h. Unsupervised learning
    i. Supervised learning
    j. Reinforcement learning
    k. No learning
    l. Unsupervised learning
    m. Unsupervised learning
    n. No learning
    o. No learning
    p. No learning
    q. Unsupervised learning
    r. Supervised learning
    s. Reinforcement learning
    t. No learning
2. Min and Argmin
    a. Min and max describe the minimum and maximum values of the list of numbers. For this example, where *l* = [-3.5, 5, -10, 0, 15.5, -2], the min is -10 and the max is 15.5. Argmin and argmax describe the input values that minimize and maximize the output values of a function, respectively. There is no function given in the example, so it's not possible to compute the argmin and argmax.
    b. It's impossible to compute the argmin and argmax without a function to evaluate the list of inputs, the function extreme_args() will need to take both a function and the list of inputs.
    c. The argmin is 0 and the argmax is 15.5.